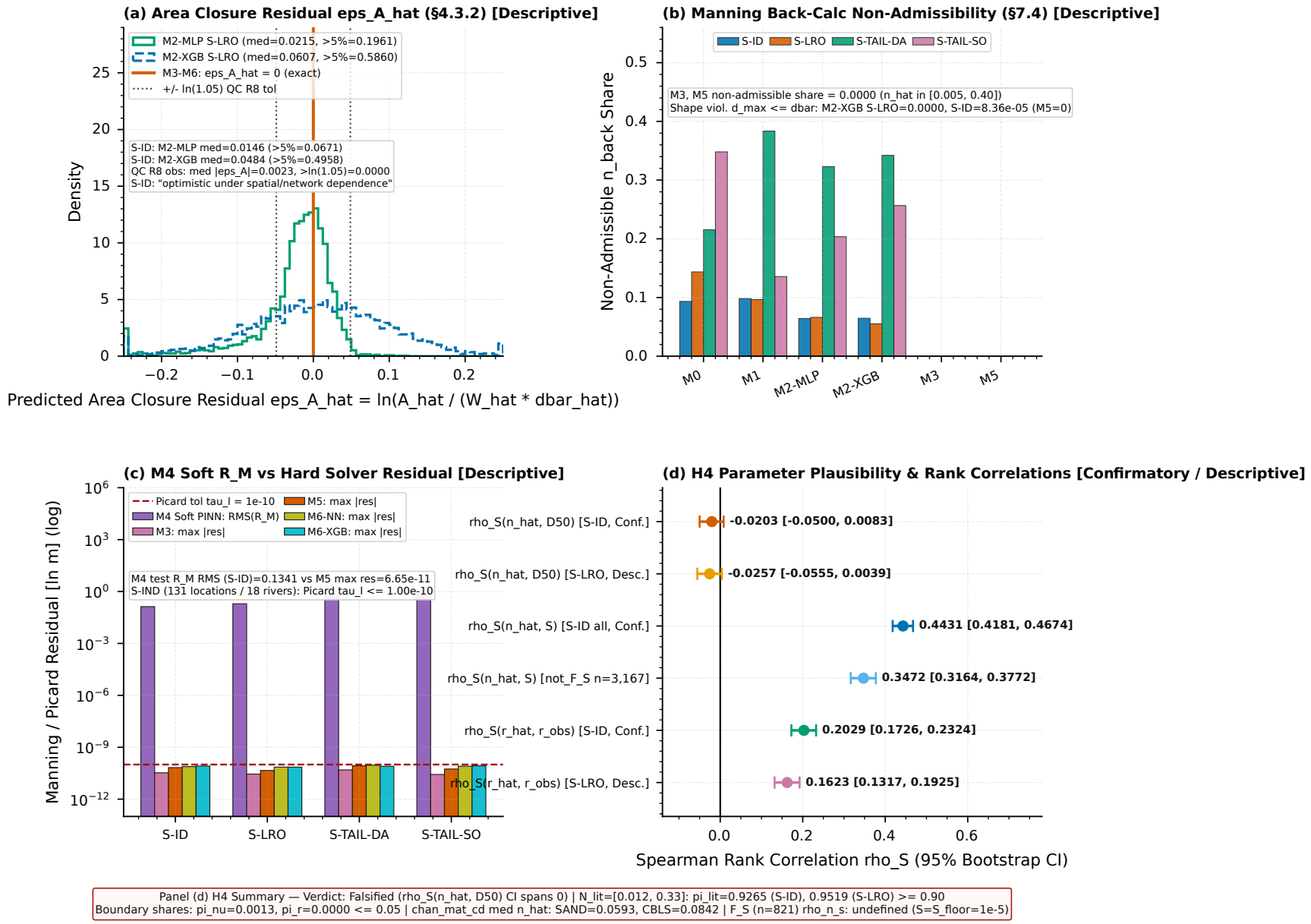
(a) Area Closure Residual eps_A_hat (§4.3.2) [Descriptive]
M2-MLP S-LRO (med=0.0215, >5%=0.1961)
M2-XGB S-LRO (med=0.0607, >5%=0.5860)
M3-M6: eps_A_hat = 0 (exact)
+/- ln(1.05) QC R8 tol
S-ID: M2-MLP med=0.0146 (>5%=0.0671)
S-ID: M2-XGB med=0.0484 (>5%=0.4958)
QC R8 obs: med |eps_A|=0.0023, >ln(1.05)=0.0000
S-ID: "optimistic under spatial/network dependence"
Density
Predicted Area Closure Residual eps_A_hat = ln(A_hat / (W_hat * dbar_hat))
(b) Manning Back-Calc Non-Admissibility (§7.4) [Descriptive]
S-ID
S-LRO
S-TAIL-DA
S-TAIL-SO
M3, M5 non-admissible share = 0.0000 (n_hat in [0.005, 0.40])
Shape viol. d_max <= dbar: M2-XGB S-LRO=0.0000, S-ID=8.36e-05 (M5=0)
Non-Admissible n_back Share
M0
M1
M2-MLP
M2-XGB
M3
M5
(c) M4 Soft R_M vs Hard Solver Residual [Descriptive]
Picard tol tau_l = 1e-10
M4 Soft PINN: RMS(R_M)
M3: max |res|
M5: max |res|
M6-NN: max |res|
M6-XGB: max |res|
M4 test R_M RMS (S-ID)=0.1341 vs M5 max res=6.65e-11
S-IND (131 locations / 18 rivers): Picard tau_l <= 1.00e-10
Manning / Picard Residual [ln m] (log)
S-ID
S-LRO
S-TAIL-DA
S-TAIL-SO
(d) H4 Parameter Plausibility & Rank Correlations [Confirmatory / Descriptive]
rho_S(n_hat, D50) [S-ID, Conf.]
-0.0203 [-0.0500, 0.0083]
rho_S(n_hat, D50) [S-LRO, Desc.]
-0.0257 [-0.0555, 0.0039]
rho_S(n_hat, S) [S-ID all, Conf.]
0.4431 [0.4181, 0.4674]
rho_S(n_hat, S) [not_F_S n=3,167]
0.3472 [0.3164, 0.3772]
rho_S(r_hat, r_obs) [S-ID, Conf.]
0.2029 [0.1726, 0.2324]
rho_S(r_hat, r_obs) [S-LRO, Desc.]
0.1623 [0.1317, 0.1925]
Spearman Rank Correlation rho_S (95% Bootstrap CI)
Panel (d) H4 Summary — Verdict: Falsified (rho_S(n_hat, D50) CI spans 0) | N_lit=[0.012, 0.33]: pi_lit=0.9265 (S-ID), 0.9519 (S-LRO) >= 0.90
Boundary shares: pi_nu=0.0013, pi_r=0.0000 <= 0.05 | chan_mat_cd med n_hat: SAND=0.0593, CBLS=0.0842 | F_S (n=821) rho_n_s: undefined (S=S_floor=1e-5)